

Risks, Rates and Ratios

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Contents

1	Summary	1
1.1	Scenarios	2
2	The binomial model	2
2.1	log-likelihood function	2
2.2	The odds	3
2.3	Standard error of the log odds	3
3	The Poisson model	4
3.1	Log-likelihood function	4
3.2	log rate parameterization	5
3.3	Poisson regression	5
4	Measures of effect	6
4.1	Risk ratio	6
4.2	Risk difference	6
4.3	Odds ratio	7
4.4	Rate ratio	7
4.5	Rate difference	7
4.6	Population attributal fraction	7
5	A profile likelihood for the rate ratios	8

1 Summary

	Exposed	Unexposed
Death	D_1	D_0
Alive	A_1	A_0
Total	n_1	n_0
Period	y_1	y_0

Parameter	Estimate ($\hat{\mu}$)	$se(\hat{\mu})^2$	$se(\log \hat{\mu})^2$
Risk	D/n	$\frac{DA}{n^3} = \frac{\hat{\mu}(1-\hat{\mu})}{n}$	—
Odds	D/A	—	$\frac{1}{D} + \frac{1}{A}$
Rate	D/y	$\frac{D}{y^2}$	$\frac{1}{D}$

Parameter	Difference ($\hat{\Delta}$)	Ratio ($\hat{\theta}$)	$se(\hat{\Delta})^2$	$se(\log \hat{\theta})^2$
Risk	RD	RR	$\frac{D_1 A_1}{n_1^3} + \frac{D_0 A_0}{n_0^3}$	$\frac{1}{D_1} + \frac{1}{n_1} + \frac{1}{D_0} + \frac{1}{n_0}$
Odds	OD	OR		$\frac{1}{D_1} + \frac{1}{A_1} + \frac{1}{D_0} + \frac{1}{A_0}$
Rate	HD	HR	$\frac{D_1}{y_1^2} + \frac{D_0}{y_0^2}$	$\frac{1}{D_1} + \frac{1}{D_0}$

1.1 Scenarios

1.1.1 Scenario 1 — significant effects

Consider a two-arm parallel study with $D_1 = 24$ events in the exposed arm and $D_2 = 42$ events in the unexposed arm. Also assume that 2400 subjects were at risk of the event in each arm for a period of 30 weeks. Though subjects with events are not at risk for the remainder of the risk period, we will ignore this aspect at take the risk time in each arm to be 195 years $((2400 - 33) \cdot 30/365)$.

Since the observation time and number of exposed subjects are the same for exposed and unexposed arms the hazard ratio and risk ratio are also the same. Further, since the outcome is rare (between 1% and 2%), the uncertainty is almost the same as well. The outcome being rare also means that the odds ratio is very similar to the risk ratio and hazard ratio; the uncertainty of the odds ratio is also close to that of the other two.

The results show a relative risk reduction of $RRR = 43\%$ (95% CI: 5% – 65%) corresponding to a hazard ratio of 0.57 (95% CI: 0.35 – 0.95).

There was an absolute risk reduction of $ARR = 0.75\%$ (95% CI: 0.09% – 1.4%) corresponding to a risk difference of $RD = -0.0075$ (95% CI: -0.014 – -0.00092) and a number needed to treat of $NNT = 133$ (95% CI: 71 – 1093).

1.1.2 Scenario 2 — non-significant effects

1.1.3 Scenario 3 — equivalence and non-inferiority

2 The binomial model

The probability mass function (PMF) for the binomial distribution gives the probability of observing $X = x$ events out of n independent trials if the probability of event in each trial is π :

$$f(x; \pi, n) = P(X = x | n, \pi) = \binom{n}{x} \pi^x (1 - \pi)^{n-x}, \quad \text{where} \quad \binom{n}{x} = \frac{n!}{x!(n-x)!}$$

A variable X having a binomial distribution is written

$$X \sim \text{binom}(\pi; n)$$

where π is the probability *parameter* and n is an *index*. Expectation and variance of the binomial distribution is

$$E[X] = n\pi, \quad \text{Var}[X] = n\pi(1 - \pi)$$

2.1 log-likelihood function

The binomial log-likelihood for an observation x can be expressed as

$$\begin{aligned} \ell(\pi; x, n) &= x \log(\pi) + (n - x) \log(1 - \pi) + \log(k), \quad \text{where} \quad k = \binom{n}{x} \\ &= n \log(1 - \pi) + x [\log(\pi) - \log(1 - \pi)] + \log(k) \\ &= n \log(1 - \pi) + x \log\left(\frac{\pi}{1 - \pi}\right) + \log(k) \end{aligned}$$

with score function

$$S(\pi; x, n) = \frac{\partial \ell(\pi; x, n)}{\partial \pi} = \frac{x}{\pi} - \frac{n-x}{1-\pi}.$$

Equating the score function to zero gives the maximum likelihood estimate (MLE)

$$\hat{\pi}(x, n) = \frac{x}{n}$$

The observed Fisher information is

$$I(\pi; x, n) = -\frac{\partial S(\pi; x, n)}{\partial \pi} = \frac{x}{\pi^2} + \frac{n-x}{(1-\pi)^2}$$

and taking its expectation gives the expected Fisher information

$$\mathcal{I}(\pi; x, n) = \mathbb{E}[I(\pi; x, n)] = \frac{n\pi}{\pi^2} + \frac{n-n\pi}{(1-\pi)^2} = \frac{n}{\pi(1-\pi)}$$

which does not depend on x . Interestingly, evaluating the observed Fisher information at the MLE gives the same expression

$$I(\hat{\pi}; x, n) = \mathcal{I}(\pi; x, n)|_{\pi=\hat{\pi}} = \frac{n}{\hat{\pi}(1-\hat{\pi})}.$$

The standard error of $\hat{\pi}$ is then

$$\text{se}(\hat{\pi}) = \mathcal{I}(\pi; x, n)^{-1/2}|_{\pi=\hat{\pi}} = \sqrt{\hat{\pi}(1-\hat{\pi})/n}$$

2.2 The odds

The odds of an event is given by

$$\Omega = \frac{\pi}{1-\pi} \quad \text{hence} \quad \pi = \frac{\Omega}{1+\Omega} \quad \text{and} \quad 1-\pi = \frac{1}{1+\Omega}.$$

The binomial log-likelihood function is sometimes reexpressed in terms of the odds rather than probabilities leading to

$$\begin{aligned} \ell(\Omega; x, n) &= x \log \left(\frac{\Omega}{1+\Omega} \right) + (n-x) \log \left(\frac{1}{1+\Omega} \right) + \text{constant} \\ &= x \log(\Omega) - n \log(1+\Omega) + \text{constant}. \end{aligned}$$

Considering now the log odds $\theta = \log(\Omega)$, the log-likelihood is

$$\ell(\theta; x, n) = x\theta - n \log(1 + e^\theta) + \text{constant}$$

2.3 Standard error of the log odds

For a scalar variable transformation $\theta = f(\pi)$ the standard error of θ is given by

$$\text{se}(\theta) = \text{se}(\pi) \left| \frac{\partial \theta}{\partial \pi} \right|$$

which can now be expressed as follows. First, recall that $\text{se}(\hat{\pi}) = \sqrt{\hat{\pi}(1-\hat{\pi})/n}$ and since

$$\theta = \log \left(\frac{\pi}{1-\pi} \right) = \log(\pi) - \log(1-\pi)$$

the derivative of θ with respect to π is

$$\frac{\partial \theta}{\partial \pi} = \frac{1}{\pi} + \frac{1}{1-\pi} = \frac{1}{\pi(1-\pi)}$$

The standard error θ is then

$$\text{se}(\theta) = \sqrt{\frac{\pi(1-\pi)}{n} \frac{1}{[\pi(1-\pi)]^2}} = \frac{1}{\sqrt{n\pi(1-\pi)}}$$

At the MLE $\hat{\theta}$ where $\hat{\pi} = x/n$ and $1 - \hat{\pi} = (n - x)/n$ this simplifies further to

$$\text{se}(\hat{\theta}) = \sqrt{\frac{1}{n} \left(\frac{1}{\hat{\pi}} + \frac{1}{1 - \hat{\pi}} \right)} = \sqrt{\frac{1}{x} + \frac{1}{n - x}}$$

3 The Poisson model

The Poisson model is a natural first model for outcomes that are counts, ie., non-negative integer numbers such as the number of traffic accidents, the number of infant deaths, the number of marriages etc., in a particular area and period. The model is parameterized by a single parameter μ , the mean number of events.

The probability mass function for the Poisson model is

$$f(x; \mu) = P(X = x | \mu) = \frac{\mu^x e^{-\mu}}{x!}$$

with expectation and variance

$$\mathbb{E}[X] = \text{Var}[X] = \mu$$

If events are observed over a time period t we can reparameterize in terms of the rate λ ; the number of events expected per unit time. The number of events during the period t is $\mu = \lambda t$, thus

$$f(x; \lambda, t) = P(X = x | \lambda, t) = \frac{(\lambda t)^x e^{-\lambda t}}{x!}$$

3.1 Log-likelihood function

The Poisson log-likelihood for an observation x is

$$\begin{aligned} \ell(\lambda; x) &= x \log(\lambda t) - \lambda t - \log(x!) \\ &= \text{constant} + x \log(\lambda) - \lambda t \end{aligned}$$

The score function is

$$S(\lambda; x) = \frac{\partial \ell(\lambda; x)}{\partial \lambda} = \frac{x}{\lambda} - t$$

giving the MLE

$$\hat{\lambda} = \frac{x}{t}$$

The observed Fisher information is

$$I(\lambda; x) = -\frac{\partial S(\lambda; x)}{\partial \lambda} = \frac{x}{\lambda^2}$$

with expectation (expected Fisher Information)

$$\mathcal{I}(\lambda; x) = \frac{t}{\lambda}$$

At the MLE the observed and expected Fisher informations are the same

$$I(\hat{\lambda}; x) = \mathcal{I}(\hat{\lambda}; x) = \frac{t^2}{x}$$

leading to the standard error $\text{se}(\hat{\lambda}) = t/\sqrt{x}$.

If a series of Poisson observations are considered $x_1, \dots, x_n$ with the same rate, then the log-likelihood is

$$\begin{aligned}\ell(\lambda, x_1, \dots, x_n) &= \sum_i [x \log(\lambda) - \lambda t + \text{constant}] \\ &= D \log(\lambda) - \lambda Y + \text{constant}\end{aligned}$$

with $D = \sum_i x_i$ and $Y = \sum_i t_i$. Since the likelihood again has the Poisson form the MLE, score and information have the same form as above but with D instead of x and Y instead of t .

3.2 log rate parameterization

It is very common to parameterize the Poisson model in terms of the log rate instead of the rate $\psi = \log \lambda$.

In terms of ψ the log-likelihood, score and Fisher information functions are

$$\begin{aligned}\ell(\psi; x) &= x\psi - te^\psi \\ S(\psi; x) &= x - te^\psi \\ I(\psi; x) &= \mathcal{I}(\psi; x) = te^\psi = t\lambda \\ I(\hat{\psi}; x) &= \frac{tx}{t} = x\end{aligned}$$

The variance of $\hat{\psi}$ is therefore

$$\text{Var}(\hat{\psi}) = \mathcal{I}(\hat{\psi}; x)^{-1} = \frac{1}{x}$$

and corresponding standard error

$$\text{se}(\hat{\psi}) = \text{se}(\log \hat{\lambda}) = \frac{1}{\sqrt{x}}$$

We could have arrived at the same result using the delta method which was applied for the odds parameter above since

$$\text{se}(\psi) = \text{se}(\lambda) \left| \frac{\partial \psi}{\partial \lambda} \right| = \sqrt{\frac{\lambda}{t}} \left| \frac{1}{\lambda} \right| = \sqrt{\frac{1}{\lambda t}}$$

then at the MLE $\hat{\lambda}t = x$, so $\text{se}(\hat{\psi}) = 1/\sqrt{x}$ as above.

A normal based (Wald) 95% confidence interval is

$$CI_{95\%} : \hat{\psi} \pm \frac{1.96}{\sqrt{x}}$$

leading to the following interval for $\hat{\lambda} = x/t$:

$$CI_{95\%} : (\hat{\lambda}/f; \hat{\lambda} \cdot f), \quad f = \exp(1.96/\sqrt{x})$$

3.3 Poisson regression

With a series of observations (y_i, t_i) , $i = 1, \dots, n$ under the assumption that each y_i has distribution

$$y_i \sim \text{Poisson}(\mu_i), \quad \mu_i = \lambda_i t_i$$

we may be interested in modeling the rate λ_i not assuming it to be constant. Describing the rate with a log-linear model $\log(\lambda_i) = x_i^\top \beta$, a log-linear model for the mean is

$$\log(\mu_i) = \log(t_i) + \log(\lambda_i) = \log(t_i) + x_i^\top \beta$$

This model can be fitted as a Poisson generalized linear model with a log-link, covariates x_i and offset $\log(t_i)$. An offset is like a covariate but where the coefficient is not estimated but fixed at 1.

Letting $z = 1$ for exposed, $z = 0$ for unexposed and $x_i^\top \beta = \alpha + z_i \beta_z$, and considering the rate ratio $\theta = \lambda_1/\lambda_0$, then

$$\log(\lambda_1/\lambda_0) = \log(\lambda_1) - \log(\lambda_0) = \alpha + 1 \cdot \beta_z - \alpha - 0 \cdot \beta_z = \beta_z$$

thus the rate-ratio is estimated as $\hat{\theta} = \exp(\hat{\beta}_z)$, the exponential of the model coefficient for the exposure covariate.

4 Measures of effect

Suppose D_0 and D_1 events are observed in two groups out of N_0 and N_1 trials respectively. Here the estimator of the probability of events in either group is $\hat{\pi}_j = D_j/N_j$ with variance $\text{Var}(\hat{\pi}_j) = \hat{\pi}_j(1 - \hat{\pi}_j)/N_j$

In the following we will be using the following relations to find the variance of γ when we know the variance of π with $\gamma = f(\pi)$:

$$\text{Var}(\gamma) = \text{Var}(\pi) \left| \frac{\partial \gamma}{\partial \pi} \right|^2$$

4.1 Risk ratio

Consider the risk ratio $RR = \pi_1/\pi_0$ with estimator

$$\widehat{RR} = \frac{D_1}{N_1} \frac{N_0}{D_0} = \frac{D_1}{D_0} \frac{N_0}{N_1}$$

Uncertainty is more symmetric on the log scale, so we consider $\log(RR) = \log(\pi_1) - \log(\pi_0)$

Considering that $\gamma = \log(\pi)$ has derivative $1/\pi$ with respect to π , the variance of $\log(RR)$ is

$$\begin{aligned} \text{Var}(\log(RR)) &= \text{Var}(\log(\pi_1)) + \text{Var}(\log(\pi_0)) \\ &= \frac{\pi_1(1 - \pi_1)}{N_1 \pi_1^2} + \frac{\pi_0(1 - \pi_0)}{N_0 \pi_0^2} \\ &= \frac{1 - \pi_1}{N_1 \pi_1} + \frac{1 - \pi_0}{N_0 \pi_0} \end{aligned}$$

At the MLE $\widehat{RR}$ we obtain

$$\begin{aligned} \text{Var}(\log(\widehat{RR})) &= \frac{N_1 - D_1}{N_1 D_1} + \frac{N_0 - D_0}{N_0 D_0} \\ &= \frac{1}{D_1} + \frac{1}{D_0} - \frac{1}{N_1} - \frac{1}{N_0} \end{aligned}$$

4.2 Risk difference

The risk difference $RD = \pi_1 - \pi_0$ has variance

$$\begin{aligned} \text{Var}(RD) &= \text{Var}(\pi_1) + \text{Var}(\pi_0) \\ &= \frac{\pi_1(1 - \pi_1)}{N_1} + \frac{\pi_0(1 - \pi_0)}{N_0} \end{aligned}$$

and at the MLE $\widehat{RD} = \hat{\pi}_1 - \hat{\pi}_0$ the variance is

$$\text{Var}(RD) = \frac{D_1(N_1 - D_1)}{N_1^3} + \frac{D_0(D_0 - N_0)}{N_0^3}$$

4.3 Odds ratio

Letting $\theta = \Omega_1/\Omega_0$ denote the odds ratio and where $\text{Var}(\log \Omega) = 1/(N\pi(1 - \pi))$ the variance of the log-odds ratio is

$$\text{Var}(\log \theta) = \frac{1}{N_1\pi_1(1 - \pi_1)} + \frac{1}{N_0\pi_0(1 - \pi_0)}$$

At the MLE $\hat{\theta}$ we have $\pi = D/N$ and $(1 - \pi) = (N - D)/N$, thus

$$\text{Var}(\log \hat{\theta}) = \frac{1}{D_1} + \frac{1}{N_1 - D_1} + \frac{1}{D_0} + \frac{1}{N_0 - D_0}$$

4.4 Rate ratio

Letting λ denote the annual number of events D observed over a period of Y years with MLE is $\hat{\lambda} = D/Y$.

Let $\theta = \lambda_1/\lambda_0$ denote the rate ratio. Because the uncertainty is more symmetric on the log scale for the rate ratio we consider $\log(\theta) = \log(\lambda_1) - \log(\lambda_0)$. First consider that the variance of the log rate $\log(\lambda)$ is

$$\text{Var}(\log(\lambda)) = \text{Var}(\lambda) \left| \frac{\partial \log(\lambda)}{\partial \lambda} \right|^2 = \frac{\lambda}{Y} \frac{1}{\lambda^2} = \frac{1}{\lambda Y}$$

The variance of the log rate ratio is then

$$\text{Var}(\log(\theta)) = \frac{1}{\lambda_1 Y_1} + \frac{1}{\lambda_0 Y_0}$$

and at the MLE we have

$$\text{Var}(\log(\hat{\theta})) = \frac{1}{D_1} + \frac{1}{D_0}$$

4.5 Rate difference

The rate difference $R_a D = \lambda_1 - \lambda_0$ where $\text{Var}(\lambda_j) = \lambda_j/Y_j$ where Y is the number of years of observation and λ is an annual rate.

The variance of the rate difference is therefore

$$\text{Var}(R_a D) = \frac{\lambda_1}{Y_1} + \frac{\lambda_0}{Y_0}$$

At the MLE $\hat{\lambda} = D/Y$, we have

$$\text{Var}(R_a D) = \frac{D_1}{Y_1^2} + \frac{D_0}{Y_0^2}$$

4.6 Population attributal fraction

The population attributal fraction (PAF) is the fraction of the risk of an outcome that can be attributed to a (causal) risk factor if eliminated completely. PAF is given by

$$\text{PAF} = \frac{P(D) - P(D^{E=0})}{P(D)}$$

where

- $P(D)$ is the population risk of the outcome (among exposed and unexposed)
- $P(D^{E=0})$ is the population risk of the outcome if all subjects in the population had been unexposed.

Both risks are marginal population risks. The latter risk will have to be adjusted for confounding if this is present.

5 A profile likelihood for the rate ratios

Consider 2 groups with D_0 and D_1 events ($D = D_0 + D_1$) observed over Y_0 and Y_1 person-years of observation respectively. The log-likelihood is the sum of contributions from each group:

$$\ell(\lambda_0, \lambda_1) = D_0 \log(\lambda_0) - \lambda_0 Y_0 + D_1 \log(\lambda_1) - \lambda_1 Y_1$$

Suppose we are interested in the rate-ratio $\theta = \lambda_1/\lambda_0$, then parameterizing in terms of λ_0 and θ the log-likelihood is

$$\begin{aligned} \ell(\lambda_0, \theta) &= D_0 \log(\lambda_0) - \lambda_0 Y_0 + D_1 \log(\lambda_0 \theta) - \lambda_0 \theta Y_1 \\ &= D \log(\lambda_0) + D_1 \log(\theta) - \lambda_0 Y_0 - \lambda_0 \theta Y_1 \end{aligned}$$

To obtain inference for θ we can develop an expression for the MLE of λ_0 and *profile* that parameter out of the log-likelihood. The resulting function is called a *profile log-likelihood*. To find the MLE of λ_0 we take the derivative of $\ell(\lambda_0, \theta)$ and equate to zero:

$$S(\lambda_0) = \frac{\partial \ell(\lambda_0, \theta)}{\partial \lambda_0} = \frac{D}{\lambda_0} - Y_0 - \theta Y_1$$

and equating to zero gives

$$\hat{\lambda}_0 = \frac{D}{Y_0 + \theta Y_1}$$

Plugging into the log-likelihood gives the profile log-likelihood for θ :

$$\begin{aligned} \ell_p(\theta) &= D \log\left(\frac{D}{Y_0 + \theta Y_1}\right) + D_1 \log(\theta) - \frac{D Y_0}{Y_0 + \theta Y_1} - \frac{\theta Y_1 D}{Y_0 + \theta Y_1} \\ &= D \log(D) - D \log(Y_0 + \theta Y_1) + D_1 \log(\theta) - \frac{D(Y_0 + \theta Y_1)}{Y_0 + \theta Y_1} \\ &= D_1 \log(\theta) - D \log(Y_0 + \theta Y_1) + D \log(D) - D \end{aligned}$$

The last terms $D \log(D) - D$ are constant with respect to θ and can therefore be ignored. Similarly we can add terms as long as they don't depend on θ and a particular trick here is to add $D_1 \log(Y_1)$ and $D_0 \log(Y_0)$ giving

$$\begin{aligned} \ell_p(\theta) &= D_1 \log(\theta) - D \log(Y_0 + \theta Y_1) + D_1 \log(Y_1) + D_0 \log(Y_0) \\ &= D_1 \log(\theta Y_1) - D_1 \log(Y_0) - D \log(Y_0 + \theta Y_1) + D_1 \log(Y_0) + D_0 \log(Y_0) \\ &= D_1 \log\left(\frac{\theta Y_1}{Y_0}\right) - D \log\left(1 + \frac{\theta Y_1}{Y_0}\right) \end{aligned}$$

Remarkably this profile log-likelihood is exactly identical to the binomial log-likelihood for the odds parameter $\Omega = \theta Y_1/Y_0$ with D_1 events out of D trials.

The odds parameter has MLE $\hat{\Omega} = D_1/(D - D_1) = D_1/D_0$ thus giving the (intuitive) MLE for the rate-ratio, the ratio of MLEs for the rates:

$$\hat{\theta} = \hat{\Omega} \frac{Y_0}{Y_1} = \frac{D_1 Y_0}{D_0 Y_1} = \frac{D_1/Y_1}{D_0/Y_0}$$

Since the standard error of log-odds parameter is well-known $\text{se}(\log \hat{\Omega}) = \sqrt{D_0^{-1} + D_1^{-1}}$ we can work out the standard error for $\log \hat{\theta}$ as

$$\text{se}(\log \theta) = \text{se}(\log \Omega) \left| \frac{\partial \log \theta}{\partial (\log \Omega)} \right| = \text{se}(\log \Omega) \cdot 1 = \sqrt{D_0^{-1} + D_1^{-1}}$$